

Sherif Gerges

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Summary

Human geneticist working at the intersection of statistical genetics, single-cell genomics, and protein structure. My research integrates large-scale genetics with structural modeling to uncover mechanisms driving neuropsychiatric disorders, with a focus on using genetics to discover novel biology for drug discovery.

Education

PhD	Harvard Medical School , Genetics and Genomics • Advisors: Mark Daly and Steve McCarroll	2017 – 2025
BS	Montclair State University , Molecular Biology; minors in Computer Science and Mathematics • Magna Cum Laude	2010 – 2013
	German University in Cairo, Egypt , Pharmaceutical Sciences (transferred)	2008 – 2010

Experience

Broad Institute of MIT and Harvard , Postdoctoral Researcher • Labs of Mark Daly and Steve McCarroll • Developing structure-based statistical methods for interpreting rare missense variation in neuropsychiatric disease, validated with cryo-EM	Cambridge, MA 2025 – present 1 year
Harvard Medical School , PhD Student • Thesis advisors: Mark Daly and Steve McCarroll • Developed a multi-omic strategy integrating large-scale single-cell transcriptomics, proteomics, and human genetics to uncover genes and pathways implicated in neuropsychiatric disorders, with emphasis on schizophrenia and autism spectrum disorders • Developed a statistical method that leverages AlphaFold3-predicted protein structures to reveal non-random mutation clustering and infer disease mechanisms at the domain level, demonstrating a powerful predictor of missense variant effects and an orthogonal tool for variant prioritization (manuscript in preparation) • Initiated and led a multi-institutional collaboration co-designing CRISPR-based cellular models and protein-based biochemical assays to test missense variants from exome studies of schizophrenia and ASD, coordinating the labs of Mark Daly, Steve McCarroll, and Hilary Finucane (Broad Institute), Poul Nissen (Aarhus University), and Jonathan Weissman (MIT) • Collaborated on a high-throughput droplet-based barcoding strategy recording RNA co-occurrence across thousands of individual synapses, revealing transcriptomic compartments that organize synaptic biology and are enriched for neuropsychiatric genetic signal (manuscript in preparation) • Collaborated on integrative genetic analyses demonstrating that variation in cis-regulatory sequences of granulocytes and monocytes accounts for a substantial proportion of tuberculosis heritability • Collaborated with the Autism Sequencing Consortium on single-cell transcriptomic expression patterns of newly identified ASD genes and variant prioritization	Boston, MA 2017 – 2025 8 years

Acurasset Therapeutics , Co-founder	2022 – present 4 years
- Co-founded with Professor Alexander Ploss, using advanced HBV infection models to identify small-molecule inhibitors of a previously unexploited step in the viral lifecycle - Hired a cross-functional team, including a contract research organization and a medicinal chemist from Harvard's Therapeutics Initiative, to advance the chemical development program - Lead all fundraising and operations; raised over \$500,000 in venture capital and non-dilutive grants	
Third Rock Ventures , Computational Consultant	2020 – 2021 1 year
- Lead computational consultant for the founding team of an early-stage venture in immuno-oncology and neuroimmune disease (now Asher Bio) - Built a statistical pipeline processing large transcriptomic datasets to identify highly specific single-cell RNA-seq cell-surface markers across indications - Work contributed to Asher's lead program, AB248, an engineered IL-2 immunotherapy selectively targeting CD8+ effector T cells	
Princeton University , Research Assistant, lab of Alexander Ploss	2014 – 2016 2 years
Molecular cloning, cell culture, and viral production to engineer hepatitis B, C, and D virus strains capable of infecting humanized mouse models	
California Institute of Technology , Research Assistant, lab of Kai Zinn	2012 – 2014 2 years
- Co-developed a stochastic block model simulating the evolution of modular biological networks, applied to yeast proteomics, mouse brain connectivity, and the C. elegans connectome - Modeled a 94-protein fat-regulation network in yeast, showing small-world (Watts-Strogatz) topology and modular organization supported by genetic and pharmacological validation	

Selected Talks

American Society of Human Genetics (ASHG)	2019-2021
Human single-cell transcriptomes identify cell types and states relevant to brain-related complex traits	
World Congress of Psychiatric Genetics (invited talk; oral talk finalist)	2021
An atlas of human cell-type transcriptomes and their roles in schizophrenia	
6th Neurological Disorders Summit (plenary talk)	2020
Human single-cell transcriptomes identify cell types and states relevant to brain-related complex traits	

Honors and Awards

Princeton University IP Accelerator Award	2023
Nucleate-Petri VC Startup Winner (\$200,000 venture funding award)	2022
John Maraganore Scientific Entrepreneurship Award (semi-finalist)	2022
Harvard Innovation Lab Award	2022
BioNJ BioPartnering Company Presentation Award	2021